



# DOM Manipulation

1. Modifying content
2. Modifying style
3. Modifying structure
4. Handling browser objects

- Once the HTML elements are accessed using different ways, content inside the selected elements can be changed using different ways. All are used for retrieving or modifying the text as per requirement.
  - using `innerText` property:
    - can return text of visible HTML nodes only
  - using `textContent` property
    - can return text of any HTML nodes including even style or script
  - using `innerHTML` property
    - can return text as well as markup within HTML nodes including child nodes
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- Selected HTML element can be styled using:
  - Using style property:
    - Style property which is an object representing the inline CSS styles of the element
    - Individual style property can be change using CSS attribute like `element.style.attribute = value`
  - Using CSS classes
    - CSS classes can be added, removed, or toggled on selected HTML elements.
    - The `classList` property returns the CSS classnames of an element.
    - This list can be used to add, remove or toggle existing classes in this list

- The HTML DOM provides methods and properties to modify the structure of an HTML document dynamically.
- This includes adding, removing, and rearranging elements and their content.
- Adding elements
  - Steps for adding new element
    - i. Create new element using createElement() method
    - ii. Set the attributes if required using setAttribute method
    - iii. Set the content(text) using textContent method
    - iv. Add this newly created element in the existing element using appendChild() method

- Removing elements
  - Steps for removing elements
    - i. Get the reference of the element using any ways of accessing
    - ii. Removing the element by moving to its parent node using `removeChild()` method
- Replacing elements
  - The `replaceChild()` method in the HTML DOM is used to replace one child node with another within a parent element which accepts first argument as new node and second argument as old node

Note: Adding, removing and replacing nodes should be often performed on the parent node

- Browser window supports few objects for managing browser related information and actions
- These objects are:
  - window
  - navigator
  - screen
  - history
  - location
- window object
  - window object serves as the global object for JavaScript code
  - Several global functions are part of the window object like alert etc
  - window.open() method in JavaScript is used to open a new window



- navigator object
  - provides information about the browser and its capabilities.
- history object
  - represents the user's navigation history for the current browser window
- screen object
  - provides information about the user's screen(monitor)
- location object
  - represents the URL of the current document